

MUHAMMAD MUSTAFA KHAN

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RESEARCH INTERESTS

Robotics and feedback control of physical systems that sense, process information, decide and act in real time. My current master's research focuses on experimental system identification, control design and real-time embedded implementation. My broader research interests include learning-enabled control and Physical AI.

EDUCATION

Gwangju Institute of Science and Technology (GIST) Sep 2025 – Expected Aug 2027
M.S., Mechanical and Robotics Engineering · CGPA 3.85 / 4.50

National University of Sciences and Technology (NUST) Sep 2019 – Jun 2023
B.E., Mechanical Engineering · CGPA 3.43 / 4.00

Selected coursework: Robot Control (in progress) · Control Engineering · Introduction to Mechatronics Design Fundamentals · AI Experience Lab · Introduction to Self-Driving Cars · Intelligent Motor Integration Project · Electronic Circuit Analysis and Signal Processing

RESEARCH EXPERIENCE

Graduate Researcher

Dynamics & Control Lab, Mechanical and Robotics Engineering, GIST Sep 2025 – present
Supervisor: Prof. Kyi Hwan Park · High-speed atomic force microscopy (AFM) control

- Measured contact-mode AFM frequency responses and identified a fourth-order structural model capturing the dominant resonance at 4.82 kHz.
- Designed Z-axis PI control by loop shaping of the measured frequency response and implemented it on an STM32F407 at 18.4 kHz, with integrator anti-windup and actuator saturation limits.
- Redesigned the AFM interface PCB with buffered level shifting and modulation-frequency amplification; integrated piezo and nanostage drive electronics and released the revised board for external fabrication.
- Assessed trace–retrace consistency on a 100 nm step standard: $r \geq 0.977$ and $\text{NRMSE} \leq 7.7\%$ up to 5 Hz; $r = 0.856$ and $\text{NRMSE} = 19.5\%$ at 20 Hz.
- Attributed the scan-rate limit to tracking bandwidth and added a digital notch at the resonance on a higher-rate STM32 build, attenuating the gain-limiting mode to raise achievable bandwidth.
- Deriving a model-predictive controller from the identified plant model and implementing it on the STM32 for real-time embedded evaluation.

Research Assistant

Human System Lab, Mechanical and Manufacturing Engineering, NUST Jun 2022 – Aug 2023
Real-time surface electromyography (sEMG) interface for post-stroke upper-limb rehabilitation

- Developed a real-time sEMG pipeline using eight Myo channels at 200 Hz, with sliding-window time-domain feature extraction and normalisation across six hand gestures.
- Compared SVM, decision tree, XGBoost, CatBoost and gradient-boosting classifiers using cross-validation; deployed the selected model for continuous online gesture inference.
- Developed one-, two- and three-DOF rehabilitation game interfaces and an evaluation protocol incorporating Fitts' law, task-completion measures and joint-angle assessment.
- Evaluated two-DOF task completion using Fitts' law: participant-level $R^2 \geq 0.98$ for ten able-bodied participants and ≥ 0.88 for two participants with chronic stroke.
- Curated and published the acquired sEMG recordings as the MUSED-I dataset on IEEE DataPort.

RESEARCH OUTPUTS

- [Poster]** M.M. Khan et al., “Model-Based Digital Proportional-Integral Control for High-Speed Contact-Mode Atomic Force Microscopy Imaging,” *23rd IFAC World Congress*, Busan, Korea, 23–28 August 2026.
- [Dataset]** M.M. Khan et al., “MUSED-I: Multi-Gesture Surface Electromyography (sEMG) Dataset for Stroke Rehabilitation,” *IEEE DataPort*, August 2023. DOI: 10.21227/04zq-yz45.

SELECTED PROJECTS

Voice-Coil Stage Design and Analog Feedback Control

GIST · Winter 2025–26

- Designed the actuator bench in SolidWorks, fabricated and assembled the hardware, and instrumented it with a PA51 power op amp and a Keyence IL-065 sensor calibrated at 0.1 V/mm.
- Modelled the stage as a mass–damper plant and characterised its open-loop frequency response.
- Designed analog PD, PI, lead and lag compensators as op-amp circuits and measured their frequency responses.
- The tested PI controller introduced substantial phase lag; selected a lead network with approximately 65° phase advance near 3 Hz for closed-loop tests.

REHABOTICS: Standalone Interactive Upper-Limb Rehabilitation System

NUST · 2022–2023

- Developed a standalone, home-based system for post-stroke upper-limb rehabilitation on a Raspberry Pi 4, combining sEMG gesture control, interactive game-based therapy and a soft pneumatic glove.
- Designed fibre-reinforced pneumatic actuators across three geometry iterations and assessed stress and deformation in the evaluated design by ANSYS structural analysis.
- Fabricated Ecoflex 00-30 actuators in 3D-printed moulds with strain-limiting fabric and fibre reinforcement, achieving 120–180° bending per finger segment.
- Implemented master–slave bilateral control so that a gesture classified from the functional hand drives pneumatic inflation and release on the paretic hand.

Induction-Motor Monitoring and Autoencoder Anomaly Detection

GIST · 2025

- Designed the data-logging architecture and the Python signal-processing and feature-extraction pipeline for an induction-motor monitoring rig.
- Acquired a healthy-operation baseline dataset from vibration, temperature, current and Hall-effect shaft-speed sensing on an Arduino.
- Implemented an autoencoder for unsupervised anomaly detection behind a real-time monitoring dashboard as a prototype.

HONOURS & AWARDS

- **Graduate Research Excellence Award – Research Assistant Scholarship**, Department of Mechanical and Robotics Engineering, GIST — December 2025.
- **NGIRI Research Funding**, Government of Pakistan — national undergraduate research grant for the final-year project, July 2023.
- **FICS International Startup Competition** — Finalist; REHABOTICS ranked 8th among 500+ teams, NUST, August 2023.

TECHNICAL SKILLS

Programming & Scientific Computing	Python (NumPy, pandas, scikit-learn) · Embedded C/C++ · MATLAB/Simulink · LabVIEW · ROS
Control & Signal Processing	System Identification · Frequency-Response Analysis · Loop Shaping · Biosignal Processing
Embedded Systems & Electronics	STM32 · Arduino · Raspberry Pi · NVIDIA Jetson Nano · Analog Signal Conditioning · PCB Design
Mechanical Design & Prototyping	SolidWorks · ANSYS Structural FEA · COMSOL · 3D Printing · Silicone Casting

ADDITIONAL QUALIFICATIONS

Online Specializations	Reinforcement Learning — University of Alberta, 2025 · Deep Learning — DeepLearning.AI, 2023 · Machine Learning — Stanford University, 2022
Languages	Urdu and Pashto (native) · English (CEFR C1) · Korean (beginner)
Standardised Tests	IELTS Academic 7.5 (October 2023) · GRE Quantitative 165/170 (September 2024)